Autonomy Engineering Portfolio

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UVM, Mechanical Engineering

UG Certificate in Robotics & Autonomy

Project code found on GitHub: https://github.com/maxwell-lucas2

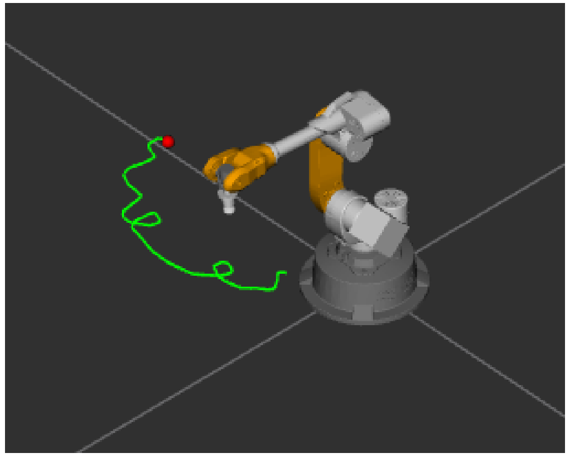
ROS2 Robotic Arm Control — WLKATA Mirobot Draw Bot

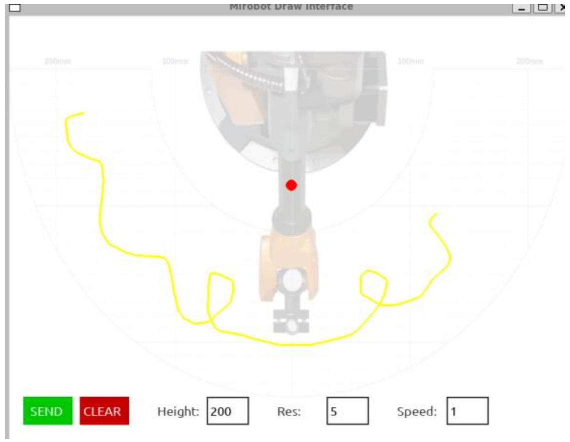
A software back-end that controls a robot to draw user-defined paths from a GUI.

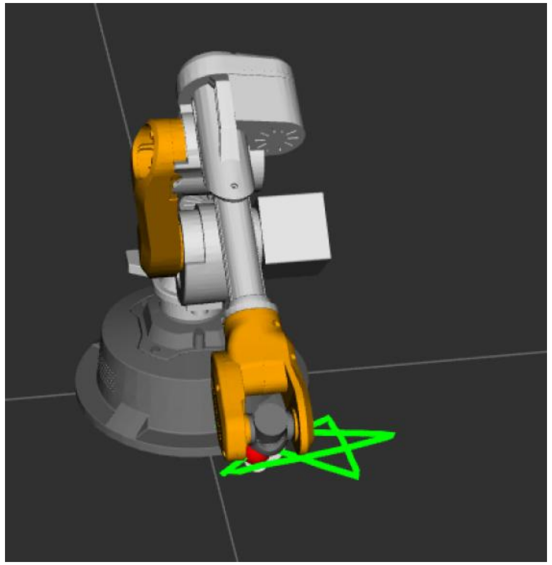
Real-time RViz2 model displaying robot pose, draw path sent through GUI node, and point on path being traced.

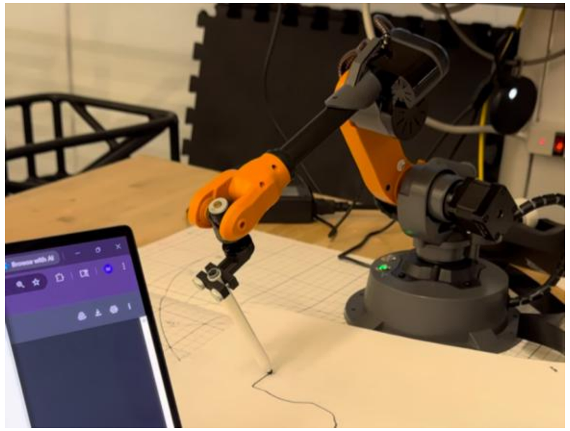
Implemented forward-kinematics through homogenous transforms, Inverse kinematics through IKPy + seed-chaining, and pure pursuit path following with variable look-ahead distance.

Independently developed as a project for the graduate level Autonomy II course at UVM.



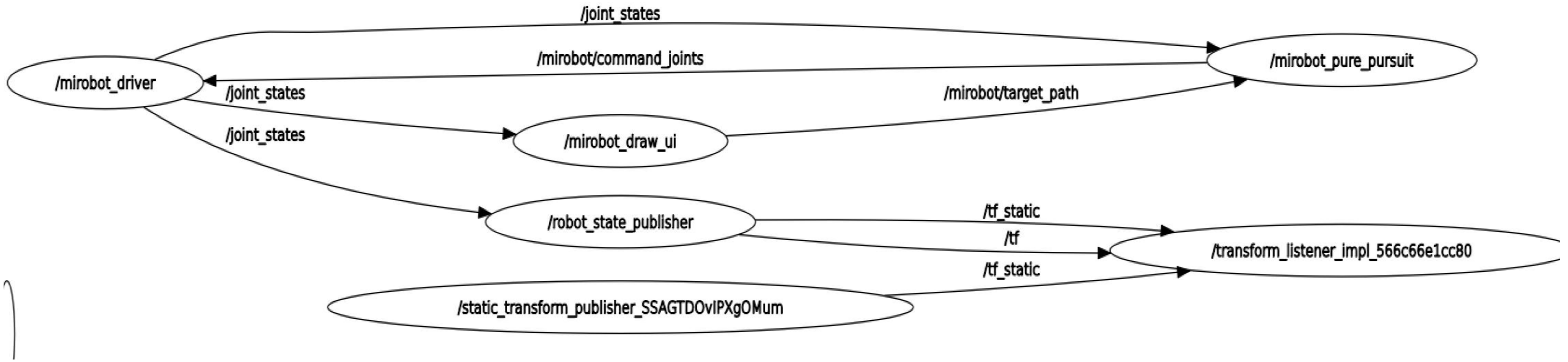






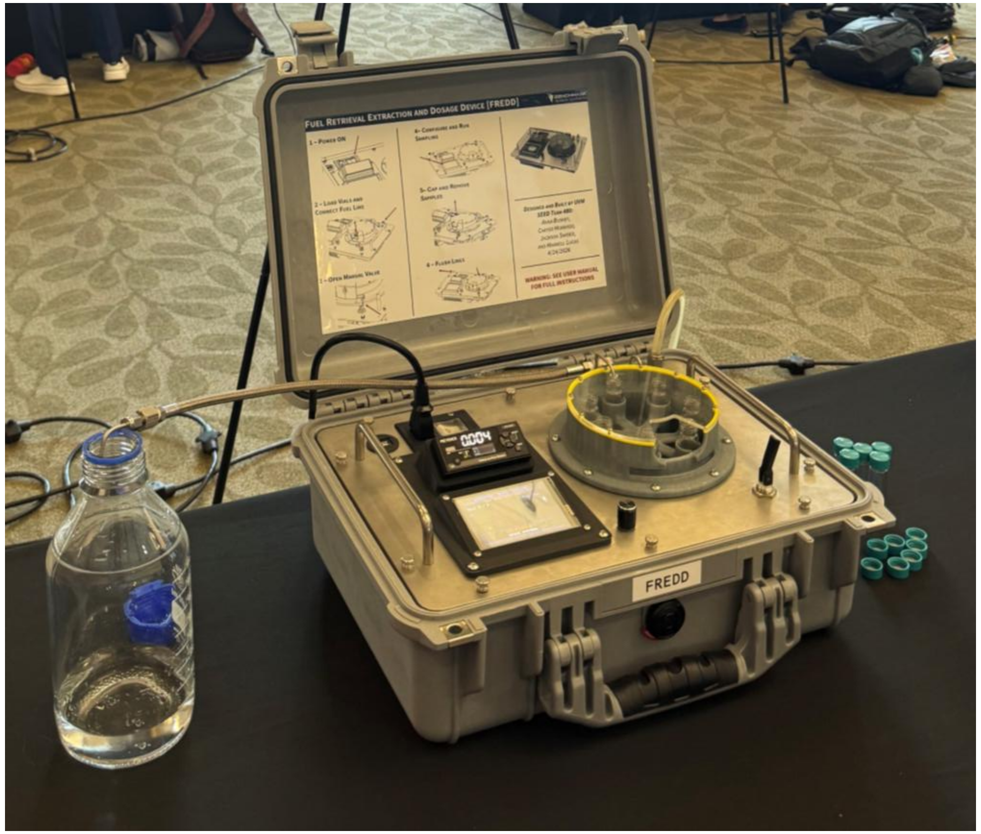
ROS2 Robotic Arm Control — WLKATA Mirobot Draw Bot

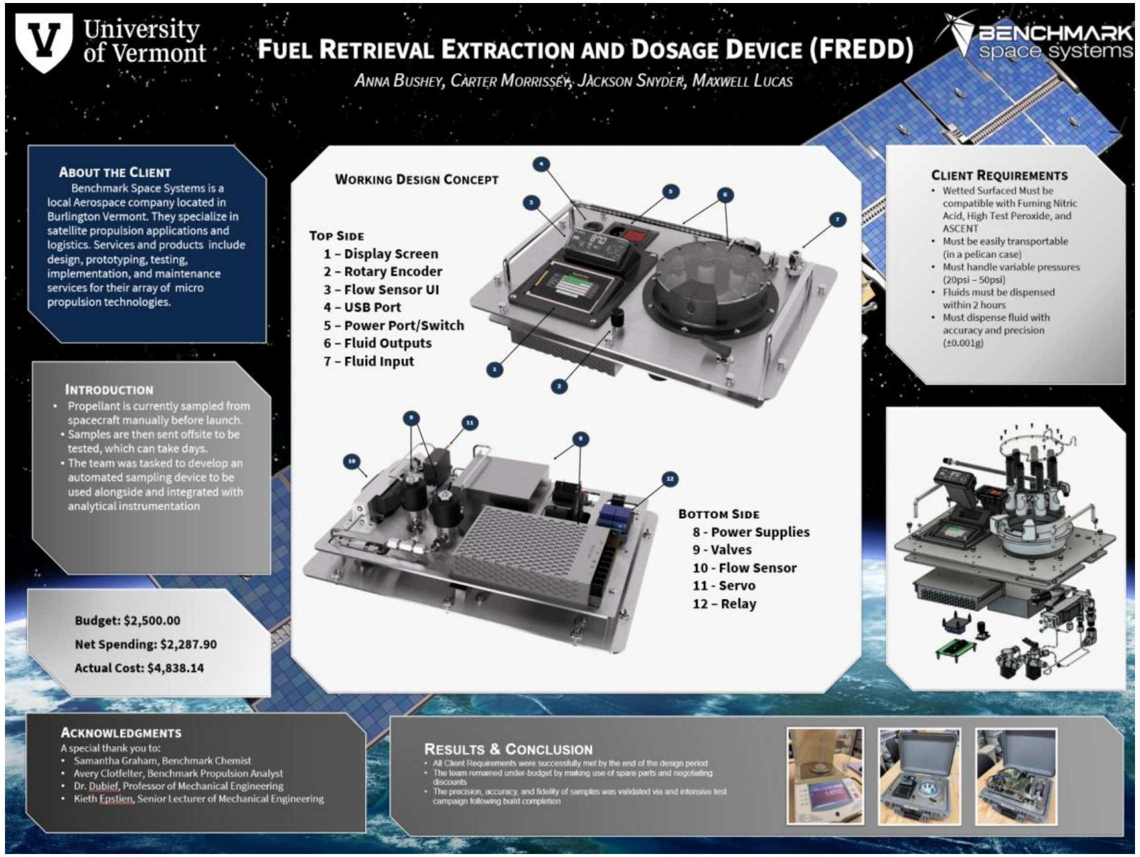
Topic Map:



Autonomous Rocket Propellant Sampling Device

Benchmark Space Systems Contract





Autonomous Rocket Propellant Sampling Device

Benchmark Space Systems Contract

Developed an aerospace grade device for Benchmark Space systems as part of a year long contract for my senior capstone.

Primarily responsible for all software and control system development, worked interdisciplinarity to develop electro-mechanical and fluid system assembly.

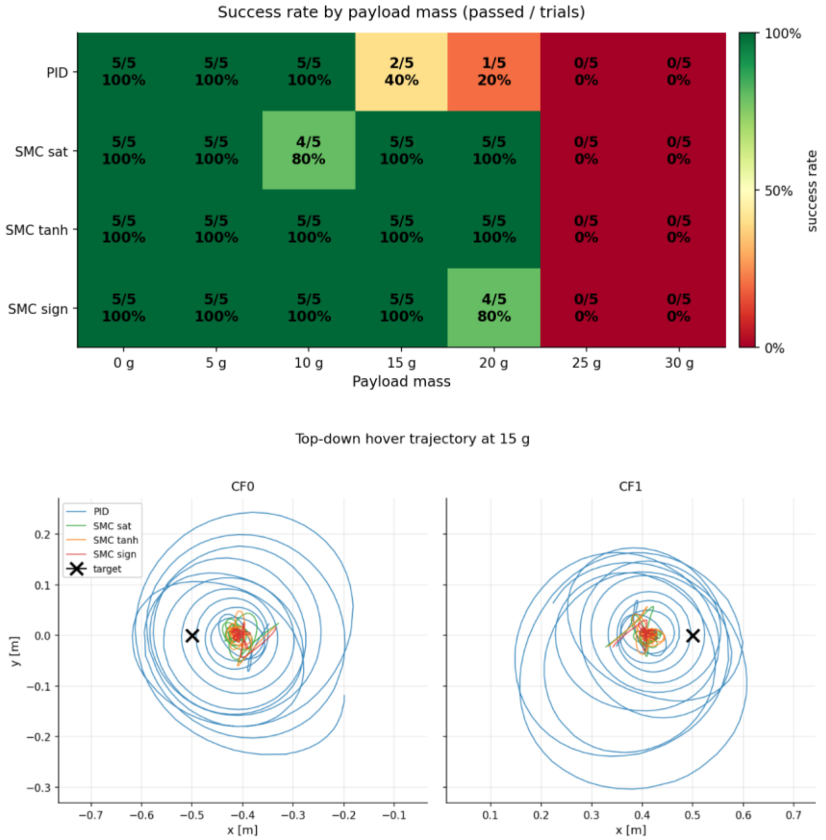
Achieved +/- 1 milligram precision over a range of 20-50 PSI agnostic to chemical properties of fluid using a high-fidelity ultrasonic volumetric flow sensor and model predicative control on a Teensy 4.1 microcontroller.

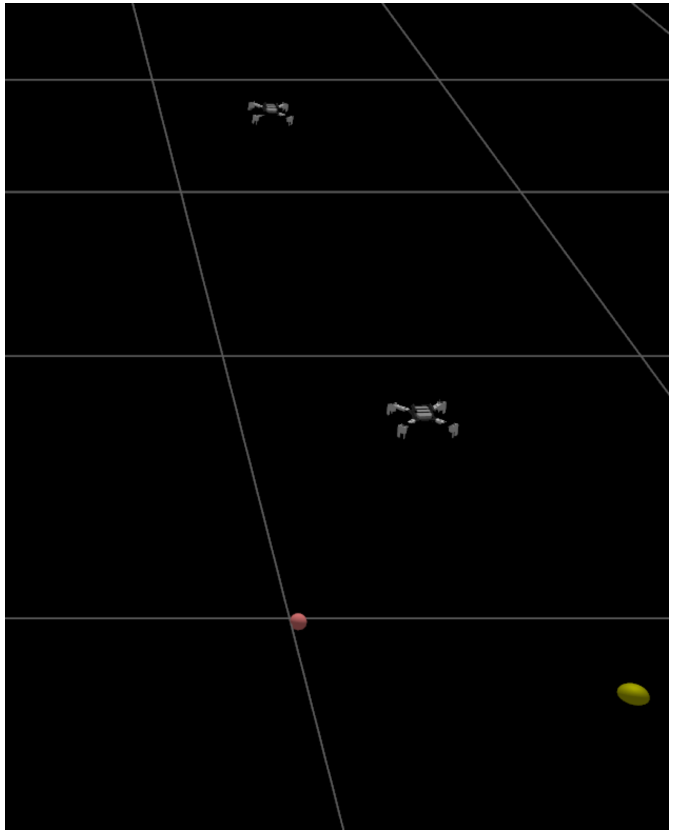
MPC strategy calculates optimal time to close valves to reach desired end mass through quadratic line fitting with an additional derivative term to achieve milligram accuracy requirements.

Awarded “SEED Production Readiness Award” for having the most complete, tested, and verified capstone project at the end of the year long period.

SMC Micro-UAV Coordinated Load Transport

Honors Undergraduate Thesis – Available at UVM ScholarWorks





SMC Micro-UAV Coordinated Load Transport

Honors Undergraduate Thesis – Available at UVM ScholarWorks

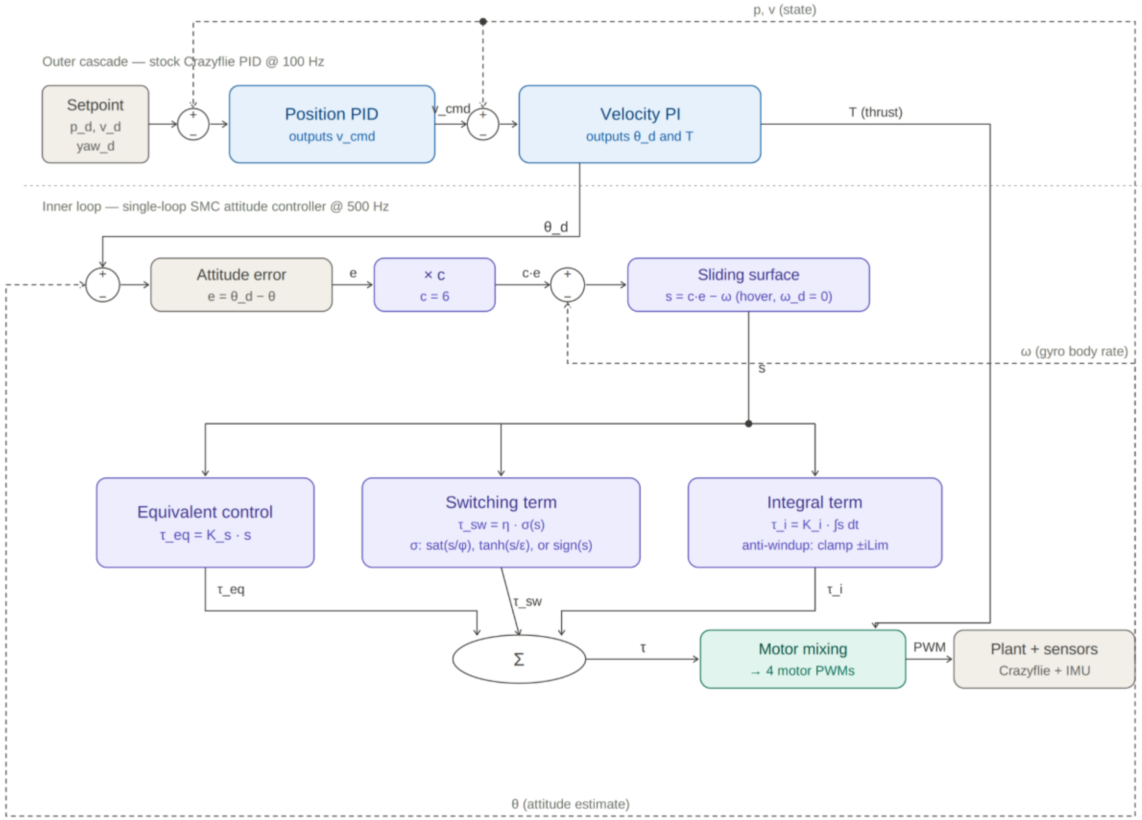
Cooperative dual-UAV payload transport using Crazyflie 2.1 platforms with spring-damper tether coupling.

Derived Lyapunov-stable SMC attitude controller; integrated with stock PID firmware as custom out-of-tree Crazyflie module.

100% SITL success rate under SMC vs. 40% for PID at 15g payload across 140 trials; ~49% reduction in peak pitch excursion.

Validated across MATLAB simulation, Gazebo Garden SITL (CrazySim), and hardware deployment.

Control System Block Diagram:



Large Electric Aircraft With Autonomous Glider Deployment

Design, Build, Fly Student Design Competition 2024-2026

Co-founded UVM Design, Build, Fly team and served as Power Systems and Autonomy Design Lead.

Developed GPS-controlled remotely deployed glider that weights 240 grams and can autonomously navigate to a destination when dropped from the larger aircraft—developed on SpeedyBee F405 Wing Mini microcontroller with Lua scripting.

Created banner-deployment and release mechanisms for 20-foot banner with a pin-release mechanism to guarantee reliable banner release.

Performed comprehensive sensitivity and trade studies in MATLAB to optimize aircraft design given mission parameters.

Performed comprehensive power-system studies and aerodynamic analysis to ensure propulsion system design would accomplish all requirements and allow for competitive performance in international DBF competition.